Monotonicity

153. Let f be a differentiable function on R such that $f(2) = 1, f'(2) = 4$. Let $\lim\limits_{x\to 0}(f(2+x))^{3/x} = e^{\alpha}$.

Then the number of times the curve $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$ meets x-axis is :

माना R पर अवकलनीय एक फलन f के लिए $f(2) = 1, f'(2) = 4$ हैं। माना $\lim\limits_{x\to 0}(f(2+x))^{3/x} = e^{\alpha}$ है। तो

वक्र $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$, x-अक्ष को कितनी बार मिलता है :

(A) 2 (B) 1 (C) 3 (D) 0

Ans. A

Sol. $\lim\limits_{x\to 0}\left(f\left(2+x\right)\right)^{\frac{3}{x}} = e^{\alpha}$ (1^{∞})

$e^{\lim\limits_{x\to 0}\left(f(2+x)-1\right)\frac{3}{x}} = e^{\alpha}$

$e^{\lim\limits_{x\to 0}\frac{f'(2+x)(3)}{1}} = e^{\alpha}$

$\Rightarrow \alpha = 12$

$\therefore y = 4x^3 - 4x^2 - 20x - 12$

$= 4(x^3 - x^2 - 5x - 3)$

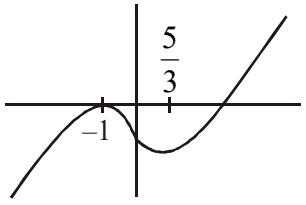
$\frac{dy}{dx} = 4\left(3x^2 - 2x - 5\right) = 0$

$3x^2 - 5x + 3x - 5 = 0$

$(x + 1)(3x - 5) = 0$

$x = -1, \frac{5}{3}$

$y(-1) = 0;\ y\left(\frac{5}{3}\right) = 4\left(\frac{125}{27} - \frac{25}{9} - \frac{25}{3} - 3\right) = -\text{ve}$



154. Let the function $f\left(x\right) = \frac{x}{3} + \frac{3}{x} + 3, x \neq 0$ be strictly increasing in $(-\infty, \alpha_1) \cup (\alpha_2, \infty)$ and strictly decreasing in $(\alpha_3, \alpha_4) \cup (\alpha_4, \alpha_5)$. Then $\sum\limits_{i=1}^{5}\alpha_i^2$ is equal to :

माना फलन $f(x)=\frac{x}{3}+\frac{3}{x}+3$, $x \neq 0$, $(-\infty,\alpha_1) \cup (\alpha_2, \infty)$ में निरंतर वर्धमान है और $(\alpha_3, \alpha_4) \cup (\alpha_4, \alpha_5)$ में निरंतर ह्रासमान है। तो $\sum_{i=1}^{5} \alpha_i^2$ बराबर है :

(A) 40 (B) 48 (C) 36 (D) 28

Ans. C

Sol. $f'(x)=-\frac{3}{x^2}+\frac{1}{3}=\frac{x^2-9}{3x^2}=\frac{(x+3)(x-3)}{3x^2}$

$\frac{+ \quad , \quad - \quad , \quad - \quad , \quad +}{-3 \qquad 0 \qquad 3}$

$f(x) \uparrow$ in $x \in (-\infty, -3) \cup (3, \infty)$

$f(x) \downarrow$ in $x \in (-3, 0) \cup (0, 3)$

$\alpha_1 = -3, \alpha_2 = 3, \alpha_3 = -3, \alpha_4 = 0, \alpha_5 = 3$

$\sum_{i=1}^{5} \alpha_i^2 = 9+9+9+0+9 = 36$

155. Let (2, 3) be the largest open interval in which the function $f(x) = 2\log_e(x-2) - x^2 + ax + 1$ is strictly increasing and (b, c) be the largest open interval, in which the function $g(x) = (x-1)^3(x+2-a)^2$ is strictly decreasing. Then $100(a+b-c)$ is equal to :

माना कि (2, 3) वह सबसे बड़ा विवृत अंतराल है जिसमें फलन $f(x) = 2\log_e(x-2) - x^2 + ax + 1$ निरंतर वर्धमान है और (b, c) वह सबसे बड़ा विवृत अंतराल है जिसमें फलन $g(x) = (x-1)^3(x+2-a)^2$ निरंतर ह्रासमान है। तब $100(a+b-c)$ बराबर है:

(A) 420 (B) 280 (C) 360 (D) 160

Ans. C

Sol. $f'(x)=\frac{2}{x-2}-2x+a \geq 0$

$f''(x)\frac{-2}{(x-2)^2}-2<0 \qquad \Rightarrow f'(x)$ is Decreasing function

$\therefore f'(3) \geq 0 \quad \Rightarrow 2-6+a \geq 0$

$a \geq 4$

Put $a = 4 \quad \Rightarrow \quad g'(x) = 3(x-1)^2(x-2)^2 + 2(x-1)^3(x-2) \leq 0$

$(x-1)^2(x-2)(5x-8) \leq 0$

$\frac{+ \quad + \quad - \quad +}{1 \qquad 8/5 \qquad 2}$

$\downarrow$ing in $\left(\frac{8}{5}, 2\right)$

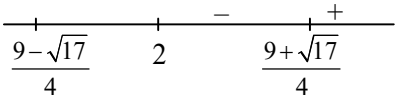
$100(a+b+c) = 10\left(4+\frac{8}{5}-2\right) = 360$

Note : If we take $a > 4$ (say $a = 5$), then

$$f'(x) = \frac{2}{x-2} - 2x + 5 \geq 0$$

$$\frac{2-2x^2+4x+5x-10}{x-2} \geq 0$$

$$\frac{2x^2-9x+8}{x-2} \leq 0$$



f(x) is ↑ing in $\left(2, \frac{9+\sqrt{17}}{4}\right)$ which is greater then (2, 3)

156. For the function $f(x) = \sin x + 3x - \frac{2}{\pi}\left(x^2+x\right)$, where $x \in \left[0, \frac{\pi}{2}\right]$ consider the following two statements :

(I) f is increasing in $\left(0, \frac{\pi}{2}\right)$

(II) f' is decreasing in $\left(0, \frac{\pi}{2}\right)$

Between the above two statements :

(A) only (I) is true

(B) only (II) is true

(C) both (I) and (II) are true

(D) neither (I) nor (II) is true

फलन

$f(x) = \sin x + 3x - \frac{2}{\pi}\left(x^2+x\right)$, जहाँ $x \in \left[0, \frac{\pi}{2}\right]$, के लिए, निम्न दो कथनों पर विचार करें।

(I) f अंतराल $\left(0, \frac{\pi}{2}\right)$ में वर्धमान है

(II) f' अंतराल $\left(0, \frac{\pi}{2}\right)$ में ह्रासमान है

उपरोक्त दो कथनों में से :

(A) केवल (I) सत्य है (B) केवल (II) सत्य है

(C) (I) तथा (II) दोनों सत्य है (D) न तो (I) और न ही (II) सत्य है

Ans. C

Sol. $f(x)=\sin x+3x-\frac{2}{\pi}(x^2+x)$, Where $x\in\left[0,\frac{\pi}{2}\right]$

$f'(x)=\cos x+3-\frac{2}{\pi}(2x+1)$

$=\cos x-\frac{4x}{\pi}-\frac{2}{\pi}+3$

as $x\in\left[0,\frac{\pi}{2}\right]$

$\frac{4x}{\pi}\in[0,2]$

$\Rightarrow 3-\frac{2}{\pi}-\frac{4x}{\pi}>0$

and also $\cos x>0$ when $x\in\left[0,\frac{\pi}{2}\right]$

$\Rightarrow f'(x)>0$

$\Rightarrow$ F(x) is increasing

Now, $f''(x)=-\sin x-\frac{4}{\pi}<0\ \forall\ x\in\left[0,\frac{\pi}{2}\right]$

Hence, f'(x) is decreasing

$\therefore$ Both statements (I) and (II) are true

157. The number of critical points of the function $f(x)=(x-2)^{\frac{2}{3}}(2x+1)$ is :

फलन $f(x)=(x-2)^{\frac{2}{3}}(2x+1)$ के क्रांतिक बिन्दुओं की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $f(x)=(x-2)^{\frac{2}{3}}(2x+1)$

$$f'(x)=2(x-2)^{\frac{2}{3}}+(2x+1)\times\frac{2}{3}(x-2)^{-\frac{1}{3}}$$

$$f'(x)=2(x-2)^{\frac{2}{3}}+\frac{2(2x+1)}{3(x-2)^{\frac{1}{3}}}$$

$$f'(x)=\frac{6(x-2)+2(2x+1)}{3(x-2)^{\frac{1}{3}}}$$

$$f'(x)=\frac{10x-10}{3(x-2)^{\frac{1}{3}}}=\frac{10(x-1)}{3(x-2)^{\frac{1}{3}}}$$

Critical points 1,2

$\Rightarrow$ 2 critical points

158. For the function $f(x) = (\cos x) - x + 1, x \in R$, between the following two statements :

(S1) $f(x) = 0$ for only one value of x in $[0, \pi]$.

(S2) f(x) is decreasing in $\left[0,\frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2},\pi\right]$.

(A) Only (S2) is correct. (B) Only (S1) is correct.

(C) Both (S1) and (S2) are correct. (D) Both (S1) and (S2) are incorrect.

फलन $f(x) = (\cos x) - x + 1, x \in R$ के लिए निचे दिये गये दो कथनों में से :

(S1) $[0, \pi]$ में x के केवल एक मान के लिए $f(x) = 0$ है

(S2) $\left[0,\frac{\pi}{2}\right]$ में f(x) ह्रासमान और $\left[\frac{\pi}{2},\pi\right]$ में वृद्धमान है

(A) केवल (S2) सत्य होगा (B) केवल (S1) सत्य होगा

(C) (S1) तथा (S2) दोनों सत्य होंगे (D) (S1) तथा (S2) दोनों असत्य होंगे

Ans. B

Sol. $f(x) = \cos x - x + 1$

$f'(x) = -\sin x - 1$

f is decreasing $\forall x \in R$

$f(x) = 0$

$f(0) = 2,\ f(\pi) = -\pi$

f is strictly decreasing in $[0, \pi]$ and $f(0).f(\pi) < 0$

$\Rightarrow$ Only one position of $f(x) = 0$

S1 is correct and S2 is incorrect.

159. Consider the function $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ defined by $f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$. Consider the statements :

(I) The curve $y = f(x)$ intersects the x-axis exactly at one point

(II) The curve $y = f(x)$ intersects the x-axis at $x = \cos\frac{\pi}{12}$.

Then :

(A) Only (I) is correct

(B) Both (I) and (II) are the correct

(C) Both (I) and (II) are incorrect

(D) Only (II) is correct

$f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$ द्वारा परिभाषित फलन $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ के लिए कथनों :

(I) वक्र $y = f(x)$, x-अक्ष को मात्र एक बिन्दु पर काटता है।

(II) वक्र $y = f(x)$, x-अक्ष को $x = \cos\frac{\pi}{12}$ पर काटता है।

में से :

(A) केवल (I) सही है

(B) (I) तथा (II) दोनों सही हैं

(C) (I) तथा (II) दोनों गलत हैं

(D) केवल (II) सही हैं

Ans. B

Sol. Let $x = \frac{\pi}{12}$

$\therefore 3x = \frac{\pi}{4}$

$\cos(3x) = \frac{1}{\sqrt{2}}$

$\Rightarrow 4\cos^3 x - 3\cos x = \frac{1}{\sqrt{2}}$

$\Rightarrow 4\sqrt{2}\cos^3 x - 3\sqrt{2}\cos x - 1 = 0$

$\therefore x = \left(\frac{\pi}{12}\right)$ is the solution of above equation statement II is correct

$f'(x) = 12\sqrt{2}x^2 - 3\sqrt{2}$

Which is positive for $x \in \left[\frac{1}{2}, 1\right]$

Hence, only one solution in $\left[\frac{1}{2}, 1\right]$

$\therefore$ Statement I is also correct

160. The function $f(x) = \frac{x}{x^2 - 6x - 16}, x \in R - \{-2, 8\}$: :

(A) decreases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(B) increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(C) decreases in (–2,8) and increases in $(-\infty, -2) \cup (8, \infty)$

(D) decreases in $(-\infty, -2)$ and increases in $(8, \infty)$

फलन $f(x) = \frac{x}{x^2 - 6x - 16}, x \in R - \{-2, 8\}$: :

(A) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में ह्रासमान है

(B) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में वर्धमान है

(C) (–2,8) में ह्रासमान है तथा $(-\infty, -2) \cup (8, \infty)$ वर्धमान है

(D) $(-\infty, -2)$ में ह्रासमान है तथा $(8, \infty)$ में वर्धमान है

Ans. A

Sol. $f(x) = \frac{x}{x^2 - 6x - 16}$

Now,

$f'(x) = \frac{-(x^2 + 16)}{(x^2 - 6x - 16)^2}$

$f'(x) < 0$

Thus f(x) is decreasing in $(-\infty,-2)\cup(-2,8)\cup(8,\infty)$

161. Let $g:R\to R$ be a non constant twice differentiable function such that $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x)=\frac{1}{2}[g(x)+g(2-x)]$, then :

(A) f" (x) = 0 for no x in (0, 1) (B) f" (x) = 0 for exactly one x in (0, 1)

(C) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$ (D) f" (x) = 0 for atleast two x in (0, 2)

माना $g:R\to R$ एक परिवर्तनीय तथा दो बार अवकलनीय फलन है और $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$ है। यदि एक वास्तविक मान फलन f, $f(x)=\frac{1}{2}[g(x)+g(2-x)]$ द्वारा परिभाषित है, तो :

(A) (0,1) में कोई भी x नहीं है, जिसके लिए f" (x) = 0 है

(B) (0,1) में ठीक एक x के लिए f" (x) = 0 है

(C) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$

(D) (0,2) में कम से कम दो x के लिए f" (x) = 0 है

Ans. D

Sol. $f'(x)=\frac{1}{2}(g'(x)-g'(2-x))$

Since $f'\left(\frac{1}{2}\right)=f'(1)=f'\left(\frac{3}{2}\right)=0$ (Rolle Theorem)

$\Rightarrow$ f"(x) = 0 for atleast 2 values of $x\in(0,2)$

162. If the function $f:(-\infty,-1]\to(a,b]$ defined by $f(x)=e^{x^3-3x+1}$ is one – one and onto, then the distance of the point P (2b + 4, a + 2) from the line $x+e^{-3}y=4$ is :

यदि $f(x)=e^{x^3-3x+1}$ द्वारा परिभाषित फलन $f:(-\infty,-1]\to(a,b]$ एकैकी तथा आच्छादी है, तो रेखा $x+e^{-3}y=4$ से बिन्दु P(2b + 4, a + 2) की दूरी है :

(A) $\sqrt{1+e^6}$ (B) $2\sqrt{1+e^6}$ (C) $4\sqrt{1+e^6}$ (D) $3\sqrt{1+e^6}$

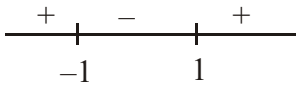
Ans. B

Sol. $f:(-\infty,-1)\to(a,b]$

$f(x)=e^{x^3-3x+1}$

$f'(x) = e^{x^3-3x+1}\left(3x^2-3\right)$

$f'(x) = e^{x^3-3x+1}3(x-1)(x+1)$



$(-\infty,-1] \rightarrow \uparrow F^n$

So $x \rightarrow -\infty$

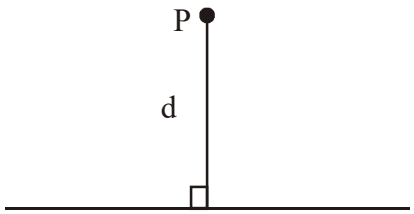
$\Rightarrow f(x) = e^{-\infty} \rightarrow 0 = a$

$x \rightarrow -1$

$\Rightarrow F(-1) = e^{-1+3+1} = e^3 = b$

$(a,b] \Rightarrow \left(0, e^3\right]$

$P\left(2e^3+4, 2\right)$



$$d = \left|\frac{2e^3+4+e^{-3}(2)-4}{\sqrt{1+\left(e^{-3}\right)^2}}\right|$$

$$d = \frac{2\left(e^3+e^{-3}\right)}{\sqrt{\frac{e^6+1}{e^6}}}$$

$$v = \frac{\frac{2}{e^3}\left(1+e^6\right)}{\frac{\sqrt{1+e^6}}{e^3}}$$

$d = 2\sqrt{1+e^6}$

163. If $5f(x)+4f\left(\frac{1}{x}\right) = x^2-2$, $\forall\, x \neq 0$ and $y = 9x^2 f(x)$, then y is strictly increasing in :

यदि $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$, $\forall\ x\neq 0$ है तथा $y = 9x^2 f(x)$ है, तो किस अन्तराल में y निरंतर वर्धमान है :

(A) $\left(0,\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$ (B) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$

(C) $\left(-\infty,\frac{1}{\sqrt{5}}\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$ (D) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

Ans. D

Sol. $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$ ...(1)

Substitute $x\to\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=\frac{1}{x^2}-2$ ...(2)

$(1)\times 5-(2)\times 4$

$\Rightarrow 9f(x)=5x^2-10-\frac{4}{x^2}+8$

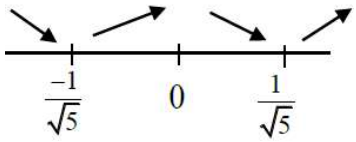
$\Rightarrow 9f(x)=\frac{5x^2-2x^2-4}{x^2}$

$\Rightarrow 9x^2f(x)=5x^4-2x^2-4$

$y=5x^4-2x^2-4$

$\frac{dy}{dx}=20x^3-4x>0$ (given)

$\Rightarrow 4x\left(5x^2-1\right)>0$



$\Rightarrow$ y is increasing in $\left(\frac{-1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

164. Let $f:[2,4]\to\mathbb{R}$ be a differentiable function such that

$(x\log_e x)f'(x)+(\log_e x)f(x)+f(x)\geq 1, x\in[2,4]$ with $f(2)=\frac{1}{2}$ and $f(4)=\frac{1}{4}$.

Consider the following two statements :

(A) : $f(x) \le 1$, for all $x \in [2, 4]$

(B) : $f(x) \ge \frac{1}{8}$, for all $x \in [2, 4]$

Then :

(A) Only statement (A) is true

(B) Both the statements (A) and (B) are true

(C) Neither statement (A) nor statement (B) is true

(D) Only statement (B) is true

माना $f : [2,4] \to \mathbb{R}$ एक अवकलनीय फलन है, जिसके लिए

$\left(x \log_e x\right) f'(x) + \left(\log_e x\right) f(x) + f(x) \ge 1, x \in [2,4]$, $f(2) = \frac{1}{2}$ तथा $f(4) = \frac{1}{4}$ है।

निम्न में दो कथनों का विचार कीजिए :

(A) : सभी $x \in [2, 4]$ के लिए $f(x) \le 1$ है।

(B) : सभी $x \in [2, 4]$ के लिए $f(x) \ge \frac{1}{8}$ है।

तो :

(A) केवल कथन (A) सत्य है

(B) (A) तथा (B) दोनों कथन सत्य हैं

(C) न तो कथन (A) न ही कथन (B) सत्य है

(D) केवल कथन (B) सत्य है

Ans. B

Sol. $\frac{d}{dx}\left(x\ \ell n\left(x\right).f(x) - x\right) \ge 0, \forall x \in [2,4]$

let $g(x) = x\ \ell n\left(x\right) f(x) - x, \quad \forall x \in [2,4]$

$\Rightarrow g'(x) \ge 0, \forall x \in [2,4]$

$\Rightarrow g'(x)$ is increasing funtion in [2,4]

$\Rightarrow g(2) \le g(x) \le g(4)$

$\Rightarrow 2 \cdot \ell n \cdot \frac{1}{2} - 2 \le x\ \ell n\left(x\right) f(x) - x \le 4 \cdot \ell n\left(4\right) \cdot \frac{1}{4} - 4$

$\Rightarrow \ln 2 - 2 + x \le x.\ell nx.f(x) \le 2(\ell n2 - 2) + x$

$\Rightarrow \frac{\ell n2 - 2}{x\ell nx} + \frac{1}{\ell nx} \le f(x) \le \frac{2(\ln 2 - 2)}{x\ell nx} + \frac{1}{\ell nx}$

Since numerator is constant

$\Rightarrow \frac{\ell n2 - 2}{4\ell n4} + \frac{1}{\ell \ln 4} \le f(x) \le \frac{2(\ell n2 - 2)}{2\ell n2} + \frac{1}{\ell n2}$

$\Rightarrow \frac{1}{8} + \frac{1}{4\ln 2} \le f(x) \le 1 - \frac{1}{\ell n2}$

$$2 \Rightarrow \frac{1}{8} < f(x) < 1, \forall x \in [2,4]$$

165. Let $f : (0, 1) \to R$ be a function defined by $f(x) = \frac{1}{1-e^{-x}}$, and $g(x) = (f(-x) - f(x))$. Consider two statements

(I) g is an increasing function in (0, 1)
(II) g is one-one in (0, 1)

Then, :

(A) Both (I) and (II) are true (B) Only (I) is true

(C) Neither (I) nor (II) is true (D) Only (II) is true

माना $f : (0, 1) \to R$, $f(x) = \frac{1}{1-e^{-x}}$ द्वारा परिभाषित है तथा $g(x) = (f(-x) - f(x))$ है। दो कथनों का विचार कीजिए।

(I) (0, 1) में g एक वर्धमान फलन है
(II) (0, 1) में g एकैकी है

तो, :

(A) (I) तथा (II) दोनों सत्य है (B) केवल (I) सत्य है

(C) न तो (I) न ही (II) सत्य है (D) केवल (II) सत्य है

Ans. A

Sol. $g(x) = f(-x) - f(x) = \frac{1}{1-e^{x}} - \frac{1}{1-e^{-x}}$

$$\Rightarrow g(x) = \frac{1+e^{x}}{1-e^{x}}$$

$$g'(x) = \frac{(1-e^{x})(e^{x}) - (1+e^{x})(-e^{x})}{(1-e^{x})^{2}}$$

$$g'(x) = \frac{2e^{x}}{(1-e^{x})^{2}} > 0$$

g (x) is increasing as well as one-one.

166. Let $f: R \to R$ be a function such that $f(x) = \frac{x^2+2x+1}{x^2+1}$. Then

(A) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

(B) $f(x)$ is one-one in $(-\infty, \infty)$

(C) $f(x)$ is many-one in $(1, \infty)$

(D) $f(x)$ is many-one in $(-\infty, -1)$

माना $f:\ R \to R$ एक फलन $f(x)=\dfrac{x^2+2x+1}{x^2+1}$ है। तब

(A) $[1,\infty)$ में $f(x)$ एकैकी है परन्तु $(-\infty,\infty)$ में एकैकी नहीं है।

(B) $(-\infty,\infty)$ में $f(x)$ एकैकी है।

(C) $(1,\infty)$ में $f(x)$ बहु–एक है।

(D) $(-\infty,-1)$ में $f(x)$ बहु–एक है।

Ans. A

Sol. $\because f(x)=\dfrac{(x+1)^2}{(x+1)}$

$$\because f'(x)=\frac{\left(x^2+1\right).2(x+1)-(x+1)^2(2x)}{\left(x^2+1\right)^2}=-\frac{2(x+1)(x-1)}{\left(x^2+1\right)^2}$$

clearly f(x) is one – one is $(-\infty,-1)$ and also in $(1,\infty)$ but f(x) is not one – one is $(-\infty,\infty)$

167. Let f and g be twice differentiable functions on R such that

$$f''(x)=g''(x)+6x,\quad f'(1)=4g'(1)-3=9,\quad f(2)=3g(2)=12$$

Then which of the following is NOT true?

(A) $g(-2)-f(-2)=20$

(B) If $-1<x<2$, then $|f(x)-g(x)|<8$

(C) There exists $x_0\in(1,3/2)$ such that $f(x_0)=g(x_0)$

(D) $|f'(x)-g'(x)|<6\Rightarrow -1<x<1$

माना फलन f तथा g, R पर दो बार अवकलनीय हैं तथा

$$f''(x)=g''(x)+6x,\quad f'(1)=4g'(1)-3=9,\quad f(2)=3g(2)=12$$

हैं। तो निम्न में से कौन सा सत्य नहीं है ?

(A) $g(-2)-f(-2)=20$ है

(B) यदि $-1<x<2$ है, तो $|f(x)-g(x)|<8$ है

(C) $x_0\in(1,3/2)$ का अस्तित्व है, जिसके लिए $f(x_0)=g(x_0)$ है

(D) $|f'(x)-g'(x)|<6\Rightarrow -1<x<1$

Ans. B

Sol. $f''(x)=g''(x)+6x$

$$\Rightarrow \int f''(x)dx=\int g''(x)dx+\int 6xdx+C$$

$$\Rightarrow f'(x)=g'(x)+3x^2+C \qquad .....(1)$$

$x = 1 \Rightarrow f'(1) = g'(1) + 3 + C$

$\Rightarrow 9 = 3 + 3 + C \Rightarrow \quad C = 3$

from (1) $\int f'(x)dx = \int g'(x)dx + \int 3x^2dx + \int 3dx + k$

$\Rightarrow f(x) = g(x) + x^3 + 3x + K$

$x = 2 \Rightarrow f(2) = g(2) + 8 + 6 + K$

$\Rightarrow 12 = 4 + 8 + 6 + K$

$\Rightarrow K = -6$

$\Rightarrow f(x) = g(x) + x^3 + 3x - 6$

Let $h(x) = f(x) - g(x) = x^3 + 3x - 6$

$\Rightarrow h'(x) = f'(x) - g'(x) = 3x^2 + 3 > 0 \ \forall \ x$

$\Rightarrow h'(x)$ is S.I. $\Rightarrow h(x) = 0$ has no solution

$\Rightarrow h(x) \in (h(-1), h(2)) \Rightarrow h(x) \in (-8, 8)$

$|f'(x) - g'(x)| < 6 \Rightarrow |3x^2 + 3| < 6 \Rightarrow 3x^2 + 3 < 6$

$\Rightarrow x^2 - 1 < 0 \Rightarrow x \in (-1, 1)$

$h(-2) = f(-2) - g(-2) = -20$

168. Let $f : R \to R$ be a thrice diffrentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeroes of $(3f'f'' + ff''')(x)$ is _____.

माना त्रिअवकलनीय एक फलन $f : R \to R$ इस प्रकार है कि $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ है, तथा $f(4) = -2$, तो $(3f'f'' + ff''')(x)$ के न्यूनतम शून्यों की संख्या है _____

Ans. 5

Sol. $\because \ f : R \to R$ and $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$ then

$f(x)$ has atleast 4 real roots.

Then $f'(x)$ has atleast 3 real roots and $f''(x)$ has atleast 2 real roots.

$$\frac{d}{dx}\left(f^3 \cdot f''\right) = 3f^2 \cdot f' \cdot f'' + f^3 \cdot f'''$$

$$= f^2\left(3f' \cdot f'' + f \cdot f'''\right)$$

Here $f^3 \cdot f''$ has atleast 6 roots.

Then its differentiation has atleast 5 distinct roots.

169. Let the set of all values of p, for which $f(x) = (p^2 - 6p + 8)(\sin^2 2x - \cos^2 2x) + 2(2 - p)x + 7$ does not have any critical point, be the interval (a, b). Then 16ab is equal to _____.

माना p के सभी मानों, जिनके लिए $f(x)=\left(p^2-6p+8\right)\left(\sin^2 2x-\cos^2 2x\right)+2(2-p)x+7$ का एक भी क्रांतिक बिन्दु नहीं हैं, का समुच्चय, अंतराल (a, b) है। तो 16ab बराबर है _____.

Ans. 252

Sol. $f(x) = -(p^2-6p+8)\cos 4n + 2(2-p)n + 7$

$f^1(x) = +4(p^2-6p+8)\sin 4x + (4-2p) \neq 0$

$$\sin 4x \neq \frac{2p-4}{4(p-4)(p-2)}$$

$$\sin 4x \neq \frac{2(p-2)}{4(p-4)(p-2)}$$

$p \neq 2$

$$\sin 4x \neq \frac{1}{2(p-4)}$$

$$\Rightarrow \left|\frac{1}{2(p-4)}\right| > 1$$

on solving we get

$$\therefore p \in \left(\frac{7}{2}, \frac{9}{2}\right)$$

Hence $a=\frac{7}{2}, b=\frac{9}{2}$

$\therefore$ 16ab = 252

170. Let $f(x) = 2^x - x^2$, $x \in R$. If m and n are respectively the number of points at which the curves $y = f(x)$ and $y = f'(x)$ intersects the x-axis, then the value of m + n is _____.

माना $f(x) = 2^x - x^2$, $x \in R$ है। यदि वक्र $y = f(x)$, $y = f'(x)$, x-अक्ष को क्रमशः m तथा n बिन्दुओं पर काटते हैं, तो m + n का मान है _____.

Ans. 5

Sol. $f(x) = 2^x - x^2$

$f(x) = 0 \quad \Rightarrow 2^x = x^2$

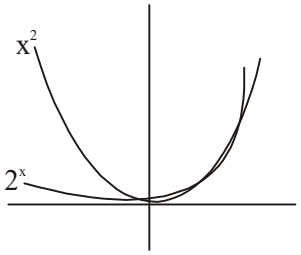
$m = 3$

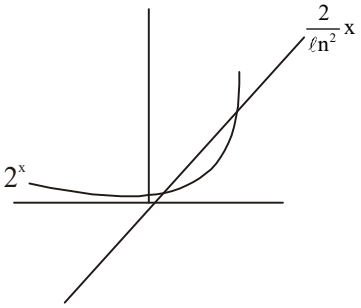
$f'(x) = 2^x \ell n2 - 2x = 0$

$2^x = \frac{2}{\ell n2} x$

$n = 2$

$m + n = 5$





171. The number of points, where the curve $y = x^5 - 20x^3 + 50x + 2$ crosses the x-axis, is _____.

उन बिंदुओं, जहाँ वक्र $y = x^5 - 20x^3 + 50x + 2$, x-अक्ष को काटता है, की संख्या है _____

Ans. 5

Sol. $y = x^5 - 20x^3 + 50x + 2$

$\frac{dy}{dx} = 5x^4 - 60x^2 + 50 = 5\left(x^4 - 12x^2 + 10\right)$

$\frac{dy}{dx} = 0 \Rightarrow x^4 - 12x^2 + 10 = 0$

$\Rightarrow x^2 = \frac{12 \pm \sqrt{144 - 40}}{2}$

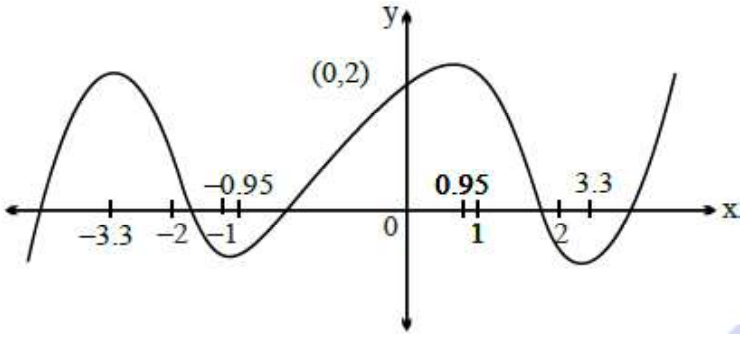
$\Rightarrow x^2 = 6 \pm \sqrt{26} \Rightarrow x^2 \approx 6 \pm 5.1$

$\Rightarrow x^2 \approx 11.1, 0.9$

$\Rightarrow x \approx \pm 3.3, \pm 0.95$

$f(0) = 2, f(1) = +\text{ ve}, f(2) = -\text{ ve}$

$f(-1) = -\text{ve}, f(-2) = +\text{ve}$



172. If a_α is the greatest term in the sequence $a_n = \frac{n^3}{n^4 + 147}, n = 1, 2, 3, \ldots$ then α is equal to _____.

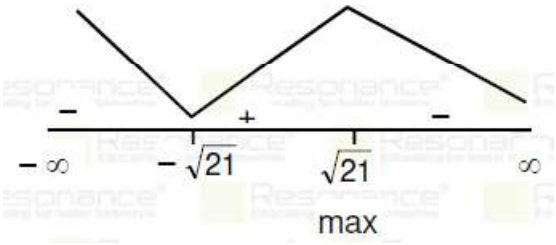
यदि अनुक्रम $a_n = \frac{n^3}{n^4+147}, n = 1,2,3,\ldots$ का अधिकतम पद a_α है, तो α बराबर है _____

Ans. 5

Sol. $f'(n) = \frac{3n^2\left(n^4+147\right) - 4n^3\left(n^3\right)}{\left(n^4+147\right)^2}$

$$f'(n) = \frac{n^2\left[3n^4+441-4n^4\right]}{\left(n^4+147\right)^2}$$

$$f'(n) = \frac{-n^2\left(n^4-441\right)}{\left(n^4+147\right)^2} = \frac{-n^2\left(n^2+21\right)\left(n^2-21\right)}{\left(n^4+147\right)^2} = \frac{-n^2\left(n^2+21\right)}{\left(n^4+147\right)^2}(n+\sqrt{21})(n-\sqrt{21})$$



When $n = 4 \quad a_4 = \frac{4^3}{4^4+147} = \frac{64}{256+147} = \frac{64}{403}$

When $n = 5 \quad a_5 = \frac{125}{625+147} = \frac{125}{772}$

a_5 is $\max = \frac{125}{772}$ hence $n = 5$

173. The number of points, where the curve $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$ cuts x-axis, is equal to _____.

उन बिंदुओं, जहाँ वक्र $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$, x-अक्ष को काटता है, की संख्या है _____

Ans. 2

Sol. $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$

$\lim\limits_{x\to-\infty} f(x) = 1, \quad f(0) = -3, \quad \lim\limits_{x\to\infty} f(x) \to \infty$

One root of f(x) = 0 is negative and other root of f(x) = 0 is positive

Curve y = f(x) cuts x-axis at two points